

RESEARCH ARTICLE

Effect of stand origin and slope position on infiltration pattern and preferential flow on a Loess hillslope

Xuemei Mei  | Qingke Zhu | Lan Ma | Dong Zhang | Yu Wang | Wenjing Hao

Key Laboratory of State Forestry
Administration on Soil and Water
Conservation, School of Soil and Water
Conservation, Beijing Forestry University,
Beijing 100083, PR China

Correspondence

L. Ma, Key Laboratory of State Forestry
Administration on Soil and Water
Conservation, School of Soil and Water
Conservation, Beijing Forestry University,
Beijing 100083, PR China.
Email: mlpcz@bjfu.edu.cn

Funding information

The National Key Research and Development
Program of China, Grant/Award Number:
2016YFC0501704; National Natural Science
Foundation of China, Grant/Award Numbers:
51779004 and 41530858

Abstract

Preferential flow is expected to provide an important pathway to replenish soil water at deep soil layers in arid or semiarid areas; however, few studies have addressed this topic, especially in semiarid Loess hillslopes. This study aimed to quantify the effect of stand origin and slope position on the contribution of preferential flow to total infiltration and spatial variations in water flow. A blue dye tracer experiment was conducted to visualize water flow in *Robinia pseudoacacia* plantation (PL), natural forestland (NF), and natural grassland (NG) at the upslope, midslope, and downslope, and semivariance analysis was used to determine spatial variability at the centimeter scale. The results showed that role of macropore flow was dominant in upslope and midslope, and larger in NF than that in PL and NG, due to presence of abundant root systems in the upper soil of NF. Moreover, contribution of preferential flow at the upper slope in NF was larger than that in PL. At the downslope, the role of rock fragments coverage at the downslope was emphasized, leading lateral flow dominant at the downslope in PL and NG. In addition, contribution of preferential flow and vertical variability of infiltration at the downslope in PL was higher than that in NF. The findings demonstrate that compared with PL, NF has a more positive impact on increasing infiltration and preferential flow that can replenish deep soil water, and reducing surface runoff and soil erosion. The presence of rock fragments coverage can make lateral flow dominant at the downslope on the Loess hillslope, related to water movement along the slope toward streams and catchment outlets.

KEYWORDS

dye tracer, infiltration pattern, Loess hillslope, preferential pathway, semivariogram

1 | INTRODUCTION

Infiltration process influences the transfer of limited rainfall amounts to the soil system affecting the availability of water to plant roots in arid and semiarid areas (Wang, Li, Xiao, Berndtsson, & Pan, 2007). Infiltration is an important factor limiting runoff generation and vegetation development (Alaoui, Caduff, Gerke, Gerke, & Weingartner, 2011). Preferential flow, which allows water to move to greater depths at faster rates, plays a critical role in infiltration process. It can significantly influence land degradation by affecting soil infiltration, regulating runoff (Allaire et al., 2011), reducing soil erosion, and increasing groundwater recharge (Bouma & De Laat, 1981).

Infiltration characteristics are influenced by vegetation type, soil physical properties (e.g., particle size distribution, initial soil water content [ISWC], and bulk density [BD]), and macropores formed by continuous root channels, soil fauna, and cracks (Gerke, 2006). Flow through macropores, called macropore flow, involves the movement of soil water along several types of preferential pathways, allowing water to bypass parts of the soil matrix. The macropore flow and lateral flow are two kinds of preferential flow. The latter is the water moving laterally (Jiang, Liu, & Zhang, 2017). Compared with preferential flow, matrix flow, which presents the other kind of soil water infiltration process, is a relatively slow and even movement of water and solutes through bulk soil (Allaire, Roulier, & Cessna, 2009). The type, extent,

and connectivity of preferential flow pathway are often assessed using dye tracer experiments (Bogner, Wolf, Schlather, & Huwe, 2008; Flury & Flühler, 1994). In a dye tracer experiment, the infiltration pattern was assessed semiquantitatively by analyzing distributions of dye coverage (DC) with depth (Flury & Flühler, 1994), by calculating the number of activated flow channels contributing to preferential flow (Gjettermann, Nielsen, Petersen, Jensen, & Hansen, 1997) or by defining categories of stained pathway widths and analyzing dye pathways of vertical and horizontal soil sections (Bachmair, Weiler, & Nutzmann, 2009; Weiler & Flühler, 2004). To date, no efforts have been made to combine the quantitative analysis of dye tracer experiment with water movement such as quantitatively analyzing the contribution of preferential infiltration to total infiltration.

Preferential flow also contributes to heterogeneity in soil water. Geostatistical methods have been used to study the spatial variability of soil water at large scales, such as the hillslope scale (Hu, Shao, Han, & Reichardt, 2011) and catchment scale (Gao, Zhao, Wu, Brocca, & Zhang, 2016). However, few efforts have focused on the spatial distribution and variability of soil water content in the presence of preferential flow at small scale. Jiang et al. (2015) studied the spatial variability in the vertical and horizontal distributions of soil water in alfalfa and corn fields by using geostatistical methods at the centimeter scale. Generally, preferential flow strongly affected spatial variation in the vertical distribution of soil water content. The infiltration pattern can reflect the presence of preferential flow in an area (van Schaik, 2009). A well-designed sampling strategy was needed using point samples to estimate infiltration pattern and the degree of preferential flow. In water-repellent, sandy soil, when collecting samples at regular intervals over a distance of a few meters to detect paths of preferential flow, the soil sample spacing should not exceed 20–25 cm (Ritsemma & Dekker, 1996). However, regardless of the sampling strategy and intervals, the possibility of the loss in full variability at the plot scale remains large (van Schaik, 2009). Therefore, based on continuous sampling methods, point samples should be collected at the centimeter scale to study the spatial variability of changes in soil water in the vertical direction and accurately infer the presence of preferential flow paths. Such an approach can improve our current understanding of potential infiltration processes and distribution characteristics.

In the Loess Plateau, new issues in ecosystems have appeared in recent years because fast growing tree and shrub species were used for vegetation restoration in “Grain for Green” Project in 1999, which requires the return of sloping cultivated land (>25 degrees) to forest and/or grass (Chen, Huang, Gong, Fu, & Huang, 2007). *Robinia pseudoacacia* is one of the most widely planted tree species in the area due to its tolerance to drought and erosion. In recent years, several negative effects of *R. pseudoacacia* have been found, such as low tree survival rate, exacerbated water shortages, and deep soil desiccation (Jiang, Cheng, Yang, & Yang, 2013). *R. pseudoacacia* strongly depletes a large amount of soil water for growth need, leading to the formation of dried layer. But it can greatly increase soil infiltrability (Laine-Kaulio, Backnas, Koivusalo, & Lauren, 2015). Thus, presence of *R. pseudoacacia* is controversial on impacting water resource, land development, and ecosystem stability in the Loess Plateau. Thus, the study in preferential flow in *R. pseudoacacia* is helpful for understanding the tree's potential impacts on water resources. In general, soil physical properties greatly

affect infiltration and preferential flow. Laine-Kaulio et al. (2015) found that in a boreal forest hillslope, lateral preferential flow was dominant at the midslope due to a high RFC and thick soil profile, whereas macropore flow was observed at the upslope and downslope, with low rock fragments and shallow soil layers. In the present study, the hypothesis that the downslope position on the Loess hillslope has a positive effect on preferential flow was proposed to improve the understanding of hydrological process due to presence of rock fragments and the relatively shallow soil depth (less than 2 m) at the downslope on the Loess hillslope.

This paper investigated the degree of preferential flow and the spatial variation of infiltration patterns by using dye tracer experiment with water movement in *R. pseudoacacia* plantation (PL), natural forestland (NF), and natural grassland (NG) at the upslope, midslope, and downslope positions, using dye tracers to determine (a) the contribution of preferential infiltration to total infiltration, (b) spatial variation of soil water infiltration, and (c) the impacts of stand origin and slope position on infiltration pattern and preferential flow, to further understand mechanism of PL impacting water infiltration.

2 | MATERIALS AND METHODS

2.1 | Experimental site

The study site was located in the Caijiachuan catchment (110°40′–110°48′E, 36°14′–36°18′N) on the southeast of Loess Plateau in Shanxi Province (Figure 1a). The area, which covers 39.33 km², is representative of the Loess hill and gully region of the Loess Plateau. The soil structure is loose and vulnerable to erosion. Gully erosion extensively distributed and intensively developed in the area. Gullies cover approximately 45% of the total area, at a density of 0.8 km·km⁻². This area has a semiarid continental climate with an average annual precipitation of 579.5 mm, average annual temperature of 10 °C, and an annual potential evaporation of 1,729 mm. The major soil type is classified as Alfisol according to the United States Department of Agriculture (USDA) classification system. On Loess hillslope, Loess soil thickness and soil structure vary with slope direction. At the slope shoulder, the soil layer is typically thick (>10 m). However, the soil containing stones is barren at the downslope, with soil depths less than 2 m and a calcic horizon below 2 m. The soil properties of the experimental site are provided in Table 1.

2.2 | Experimental process

To evaluate the effects of stand origin and slope position on infiltration pattern and preferential flow, three adjacent slopes of different vegetation types (PL, NF, and NG) with similar geomorphological characteristics and two replicated transects with three different slope positions in each slope were selected (Figure 1b–d). The two replicated transects have similar characteristics in vegetation, soil, and terrain. In each transect, three sample sites are established at the upslope, midslope, and downslope, respectively. The sample sites were located at least 2 m apart to avoid interference. In total, 18 dye experiments were performed. The PL site was planted with *R. pseudoacacia* in 2002 and had been converted from NG where land degradation such as intensive soil desiccation has taken place. The PL plot was located between two

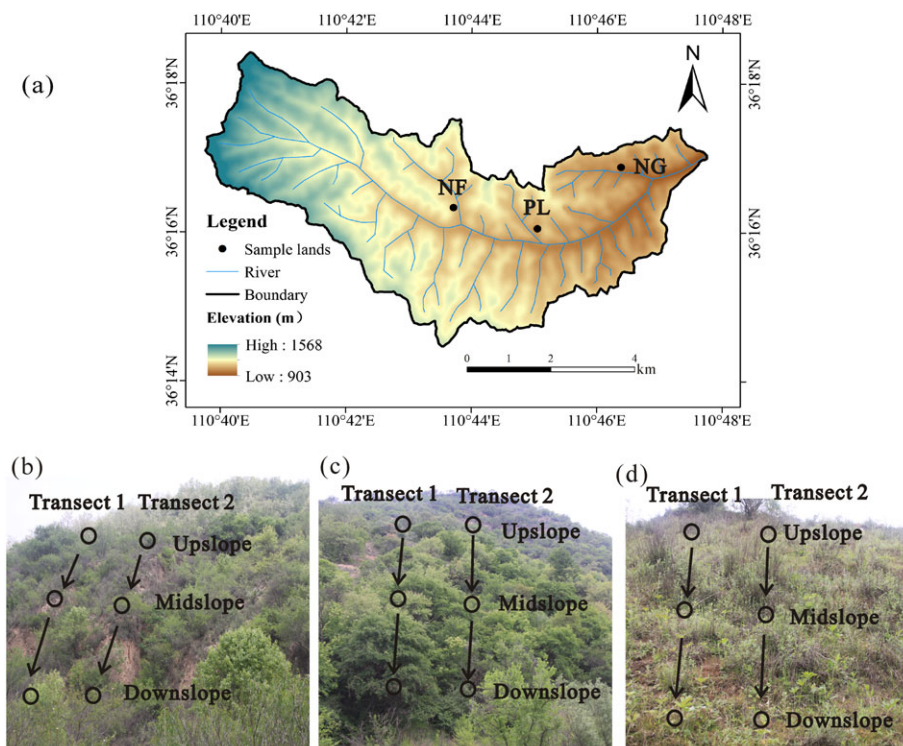


FIGURE 1 (a) Location of the research site (Caijiachuan catchment); (b), (c), (d) represent the plantation, natural forestland, and natural grassland slopes, respectively [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Mean soil physical properties in 0- to 40-cm layer

Plot	Slope position	ISWC (100 g·g ⁻¹)	BD (g·cm ⁻³)	Ks (mm·min ⁻¹)	RFC (cm ² ·cm ⁻²)	Particle size		
						Clay (%; <0.002 mm)	Silt (%; 0.002–0.02 mm)	Sand (%; 0.02–2 mm)
PL	Upslope	7.51 ± 0.18b	1.26 ± 0.05b	0.56 ± 0.01b	0.00	2.24 ± 0.17a	49.99 ± 2.42a	47.76 ± 2.59b
	Midslope	8.12 ± 0.78b	1.25 ± 0.08ab	1.18 ± 0.03b	0.00	1.88 ± 0.20ab	46.56 ± 2.60ab	51.55 ± 2.79ab
	Downslope	8.31 ± 0.04c	1.23 ± 0.08b	2.53 ± 0.04b	0.05	1.66 ± 0.16b	38.38 ± 1.64a	59.95 ± 1.80a
NF	Upslope	7.69 ± 0.44b	1.18 ± 0.05b	0.64 ± 0.01b	0.00	2.06 ± 0.14b	42.65 ± 1.85b	55.28 ± 1.86a
	Midslope	9.09 ± 0.37ab	1.17 ± 0.06b	2.72 ± 0.05a	0.00	1.47 ± 0.18b	37.99 ± 2.18b	60.54 ± 2.36a
	Downslope	10.98 ± 1.29b	1.17 ± 0.08c	3.64 ± 0.14b	0.00	1.93 ± 0.20b	39.36 ± 3.00a	58.71 ± 2.86a
NG	Upslope	8.60 ± 0.38a	1.34 ± 0.04a	1.16 ± 0.03a	0.00	2.05 ± 0.60b	51.35 ± 3.61a	46.58 ± 4.20b
	Midslope	9.29 ± 0.87a	1.33 ± 0.05a	1.93 ± 0.07ab	0.00	2.29 ± 0.24a	52.01 ± 3.56a	45.69 ± 3.80b
	Downslope	12.11 ± 1.18a	1.34 ± 0.06a	7.24 ± 1.32a	0.08	2.13 ± 0.24a	42.11 ± 2.85a	55.75 ± 3.08a

Note. Data represent means and standard deviations (SD). Different lowercase letters indicate significant differences among slope positions by vegetation type ($p < .05$). BD = bulk density; ISWC = initial soil water content; Ks = soil saturated hydraulic conductivity; NF = natural forestland; NG = natural grassland; PL = plantation; RFC = rock fragments coverage.

trees and beneath trees' canopy. The NF plot was beneath a clump of trees. The grass vegetation was cut ahead of the sprinkling in NG sites. The basic descriptions of the experimental sites are provided in Table 2. The experiments were conducted on in August 2016, 2 weeks after the latest rainfall. The two replicated transects in the hillslope for dye tracer experiments were available, as in studies of Zhang, Niu, Yu, Zhu, and Du (2015) and Zhang, Niu, Zhang, and Zhu (2017) selecting two replicates in three vegetation types for preferential flow paths, and Holden and Burt (2003) choosing two plots at each slope position in the hillslope to measure runoff and identify impact of preferential flow paths on spatial variation on runoff patterns.

Soil properties (e.g., BD, soil saturated hydraulic conductivity (Ks), ISWC, rock fragments coverage (RFC), clay, silt, and sand) prior to dye tracer application were determined in adjacent area to dye site (Jiang

et al., 2017). Due to the silt loam soils with homogeneous soil texture in the Chinese Loess Plateau, soil properties in adjacent area are coherent with that in dye site. At each sampling site, ISWC, particle size, BD, and Ks in each site were measured at the soil layer (0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm) with three replicates. ISWC was measured with oven-drying method (at 105 °C for 24 hr). The air-dried soil sample passed through a 2-mm sieve; after which, clay (<0.002 mm), silt (0.002–0.05 mm), and sand (0.05–2 mm) were measured using a Malvern Mastersizer 2000 laser diffraction device (Malvern Instruments Ltd., Malvern, UK). Three replicates of undisturbed soil cores were obtained by ring knife (diameter of 5 cm, length of 5 cm) at every 10 cm in the 0- to 40-cm soil layer to measure the BD (Wang, Wang, Wei, Shao, & Li, 2008). And stainless steel cylinders (5 cm high and 7-cm diameter) with three replicates at every 10 cm

TABLE 2 Basic description of the experimental sites

Site	Slope position	Slope aspect (°)	Land coverage (%)	Main plant type	Stand density (m)	Tree height (m)	Root biomass at the soil depth of 0–40 cm in the PVC quadrat
PL	Upslope	203	70	<i>Robinia pseudoacacia</i>	2 × 2.5	7.6	<2 mm in diameter: 9.37 g; 2–5 mm: 7.82 g; >5 mm: 84.53 g
	Midslope	203	75	<i>Robinia pseudoacacia</i>	2 × 2.5	8.3	<2 mm in diameter: 6.02 g; 2–5 mm: 6.75 g; >5 mm: 113.75 g
	Downslope	203	65	<i>Robinia pseudoacacia</i>	2 × 2.5	7.8	<2 mm in diameter: 5.46 g; 2–5 mm: 10.43 g; >5 mm: 122.61 g
NF	Upslope	103	65	Dominant trees: <i>Syringa reticulata</i> ; Companion trees: <i>Acer buergerianum</i> Shrub: <i>Rosa xanthina</i> , <i>Spiraea trilobata</i>	4 × 6	5.5	<2 mm in diameter: 30.71 g; 2–5 mm: 18.82 g; >5 mm: 127.53 g
	Midslope	103	65	Dominant trees: <i>Acer buergerianum</i> ; Companion trees: <i>Syringa reticulata</i> ; <i>Populus davidiana</i> Shrub: <i>Spiraea trilobata</i> , <i>Ostryopsis davidiana</i>	4 × 6	5.7	<2 mm in diameter: 27.64 g; 2–5 mm: 21.58 g; >5 mm: 142.43 g
	Downslope	103	70	Dominant trees: <i>Populus davidiana</i> ; Companion trees: <i>Syringa reticulata</i> Shrub: <i>Amygdalus davidiana</i> , <i>Ostryopsis davidiana</i>	4 × 6	7.1	<2 mm in diameter: 24.67 g; 2–5 mm: 29.53 g; >5 mm: 152.34 g
					3 × 4	2.1	
NG	Upslope	333	80	<i>Carex humilis</i> Leyss, <i>Artemisia lavandulae folia</i> , and <i>Bothriochloa ischcemum</i>	–	–	<2 mm in diameter: 23.31 g; 2–5 mm: 2.79 g; >5 mm: 0.28 g
	Midslope	333	85	<i>Carex humilis</i> Leyss, <i>Artemisia lavandulae folia</i> , and <i>Bothriochloa ischcemum</i>	–	–	<2 mm in diameter: 17.33 g; 2–5 mm: 3.88 g; >5 mm: 0.19 g
	Downslope	333	70	<i>Carex humilis</i> Leyss, <i>Artemisia lavandulae folia</i> , and <i>Bothriochloa ischcemum</i>	–	–	<2 mm in diameter: 15.41 g; 2–5 mm: 3.01 g; >5 mm: 0.32 g

Note. NF = natural forestland; NG = natural grassland; PL = plantation.

in the 0- to 40-cm soil layer were used to collect soil cores for measuring Ks using the constant head method (Klute & Dirksen, 1986). RFC, proportional area of rock fragments resting on the soil vertical profile, was easily observed using drawn and grid methods. Root biomass is obtained by roots dried for 48 hr at 70 °C and then weighed. The roots in PVC quadrat at 0- to 40-cm soil depth were collected at the end of the experiment.

At each site, a 30-cm-high quadrat made of PVC (inner diameter = 38 cm) was embedded 10 cm into the soil in a flat area where surface freshly fallen leaves had been removed, leaving behind an undisturbed litter layer with some older leaves on top. The same sprinkling experiment was conducted with a field sprinkler with an intensity of 15 mm·hr⁻¹ at each site (Hagedorn & Bundt, 2002). Ten liters of Brilliant Blue FCF dye solution (approximately 88-mm rainfall) with a concentration of 4 g·L⁻¹ (Alaoui et al., 2011) were uniformly applied to the surface area of PVC quadrat. The infiltration time, that is, the time elapsed when the solution disappeared, was recorded. Then each PVC quadrat was covered with a waterproof cloth to avoid solution evaporation. The waterproof cloth and PVC were removed 24 hr after the end of infiltration. A vertical soil profile was excavated at approximately the center of the quadrat depending on the dye's penetration depth. Earthworms, ants, and other burrowing animals were not observed during the excavation. A graded frame was placed on the pits' soil walls before photographing to provide references for image analysis. The soil profiles in the vertical direction were photographed with a Canon digital camera under daylight conditions beneath a black umbrella to provide a soft light.

Soil samples were collected for 18 vertical profiles after the images were obtained. Each vertical profile was subdivided into many

small grids of 2 × 2 cm (height × width) to extract soil samples with volume of 4 cm³ (2 × 2 × 1 cm [height × width × thickness]; Figure 2a). As samples were collected within the maximum infiltration depth in each vertical profile, a total of at least 190 soil samples were collected in the vertical dimension (width 38.0 cm, greatest depth 20.0 cm), and at most, a total of 361 soil samples were collected (width 38.0 cm, greatest depth 38.0 cm). Measurements of gravimetric soil water content were performed for all of the samples using the oven-drying method (at 105 °C for 24 hr) and used to calculate infiltration volume.

2.3 | Data analysis

Image analysis was geometric corrected by using ERDAS Image 9.2 software (Cey, 2007) and then classified into dye stained and nonstained classes (Wu et al., 2015). Black and white zones represented the stained and unstained areas, respectively, shown in Figure 2. Finally, the images were stored as bitmap format with a resolution of 100 pixels per 1 cm². Preferential flow indices were calculated from the dye images using Matlab7.0 software (see Figure 2 for visual examples) and included (a) DC (%), defined as the percent coverage of the stained areas (Flury & Flühler, 1994), and (b) preferential flow fraction (PF – fr [%]) described by van Schaik (2009) and calculated as the fraction of preferential flow area to the sum of the dye-stained areas as follows:

$$PF-fr = 100 \cdot \left(1 - \frac{UniFr \cdot W}{TSA} \right), \quad (1)$$

where *UniFr* is the uniform infiltration depth (cm) calculated as the depth at which DC is 80%, *W* indicates profile width (38 cm in our study), and *TSA* denotes the total stained area (cm²). The length index

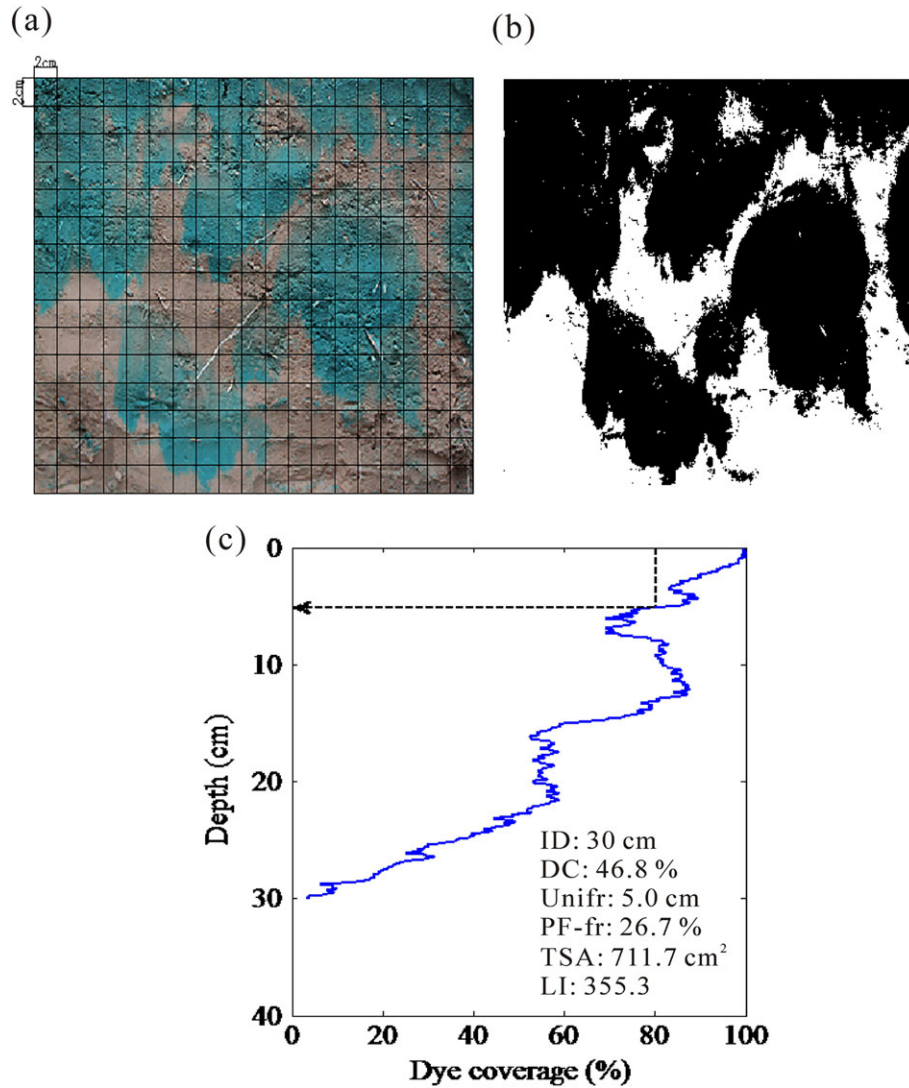


FIGURE 2 Example of stained profiles at downslope in plantation and sampling style (a), corresponding binary digital image (b), and dye coverage with depth and diagram parameters (c) [Colour figure can be viewed at wileyonlinelibrary.com]

(LI) is defined as sum of the absolute differences between consequent DC values with depth in a profile (Bargués Tobella et al., 2014).

Additionally, preferential infiltration volume (*PIV*) and its contribution to total infiltration volume were also considered. *PIV* represents the amount of infiltration corresponding to the preferential flow depth and was calculated as follows:

$$PIV = \sum_{i=0}^{i=ID} V_i - \sum_{j=0}^{j=Unifr} V_j, \quad (2)$$

where *ID* (cm) is infiltration depth and V_i , V_j (mm) indicates the amount of infiltration volume at *i*, *j* soil layer, calculated as follows:

$$V = 10 \sum_{u=1}^{u=19} (\theta_{tu} - \theta_{iu}) \cdot d \cdot v_s / 38, \quad (3)$$

where θ_{tu} is the gravimetric soil moisture content after irrigation (%), θ_{iu} is the initial gravimetric soil moisture content (%), *u* represents the width of the soil sample sequence (the maximum value was 19 as soil was sampled at 2-cm intervals within each 38-cm-wide quadrat), *d* is the soil BD of the layer ($\text{g}\cdot\text{cm}^{-3}$), and v_s is the soil volume (cm^3), which

was 4 cm^3 in this study. Contribution of preferential infiltration (%) can be defined as the proportion of preferential infiltration (*PIV*) to total infiltration volume (*TIV*), calculated as follows:

$$\text{Contribution} = 100 \cdot \frac{PIV}{TIV}. \quad (4)$$

The spatial variability of water infiltration content was quantified using semivariograms, which were described as a function of the distance between sampling points and calculated as

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z_i - Z_{i+h}]^2, \quad (5)$$

where Z_i and Z_{i+h} is measured variable at location *i* and *i* + *h*, respectively; *h* is the lag distance; and *N*(*h*) is the number of pairs separated by *h*. Four theoretical models (spherical, exponential, linear, and Gaussian) were chosen to describe semivariograms, and an optimal theoretical model with the smallest residual sum of squares and the largest coefficient of determination (R^2) were selected. Then spatial variance can be derived by the three geostatistical parameters (nugget, sill, and range).

Nugget represents microheterogeneity or experimental error. Sill represents the variance of the random soil moisture field. And the range is the distance beyond which the data are no longer autocorrelated. The ratio between nugget and sill can be used as a criterion for classifying the spatial dependence of soil water infiltration content. A variable is considered to have strong, moderate, or weak spatial dependence if nugget/sill ratios are <0.25 , 0.25 to 0.75 , or >0.75 , respectively. Ordinary kriging was used for distribution maps for soil water infiltration at the various depths. All the geostatistical computations were conducted using GS+ 7.0 (Gamma Design Software LLC., Plainwell, MI, USA). Pearson correlation coefficient analysis was used to test the correlations between environmental variables and each of infiltration and preferential flow by using SPSS 18.0 software. The differences of soil physical properties in different vegetation types were compared using one-way analysis of variance using SPSS 18.0 software.

3 | RESULTS

3.1 | Contribution of preferential flow

Eighteen stained images from vertical soil sections in the three hillslopes were investigated, and nine examples are shown in

Figure 3. In the three vegetation types, the dye was distributed consistently throughout the area in longitudinal soil sections at the upslope and midslope; however, at the downslope, the distribution of the dye was not uniform as evidenced by an unstained fraction within the staining area (Figure 3). In general, macropore flow in the vertical direction was the dominant flow behavior at the upslope and midslope. In contrast, preferential flow, which includes macropore flow in all directions and lateral flow, was the dominant flow type at the downslope.

Along with the preferential flow variables (Table 3), infiltration time, ID, DC, PF-fr, LI, PIV, and contribution showed consistent trends with slope position that increased from upslope to downslope in PL and NG. In contrast, the relationship between UniFr and slope position showed the opposite trend. But no obvious relationship between preferential flow variables and slope position was observed in NF. Extent of preferential flow was the highest at the downslope in PL and NG slopes, but not in the NF slope. Among the three vegetation types, NG took 20.47–120.01 min to infiltrate with 10-L water, longer than PL (16.62–45.57 min) and NF (20.60–24.90 min). The longer infiltration duration reflected a smaller infiltration rate, therefore, the slowest infiltration rate was observed in NG. Contribution of preferential flow at the downslope in PL (56.64%, 74.32% in Transect 1, Transect 2, respectively) and NG (52.26%, 75.21% in Transect 1, Transect 2,

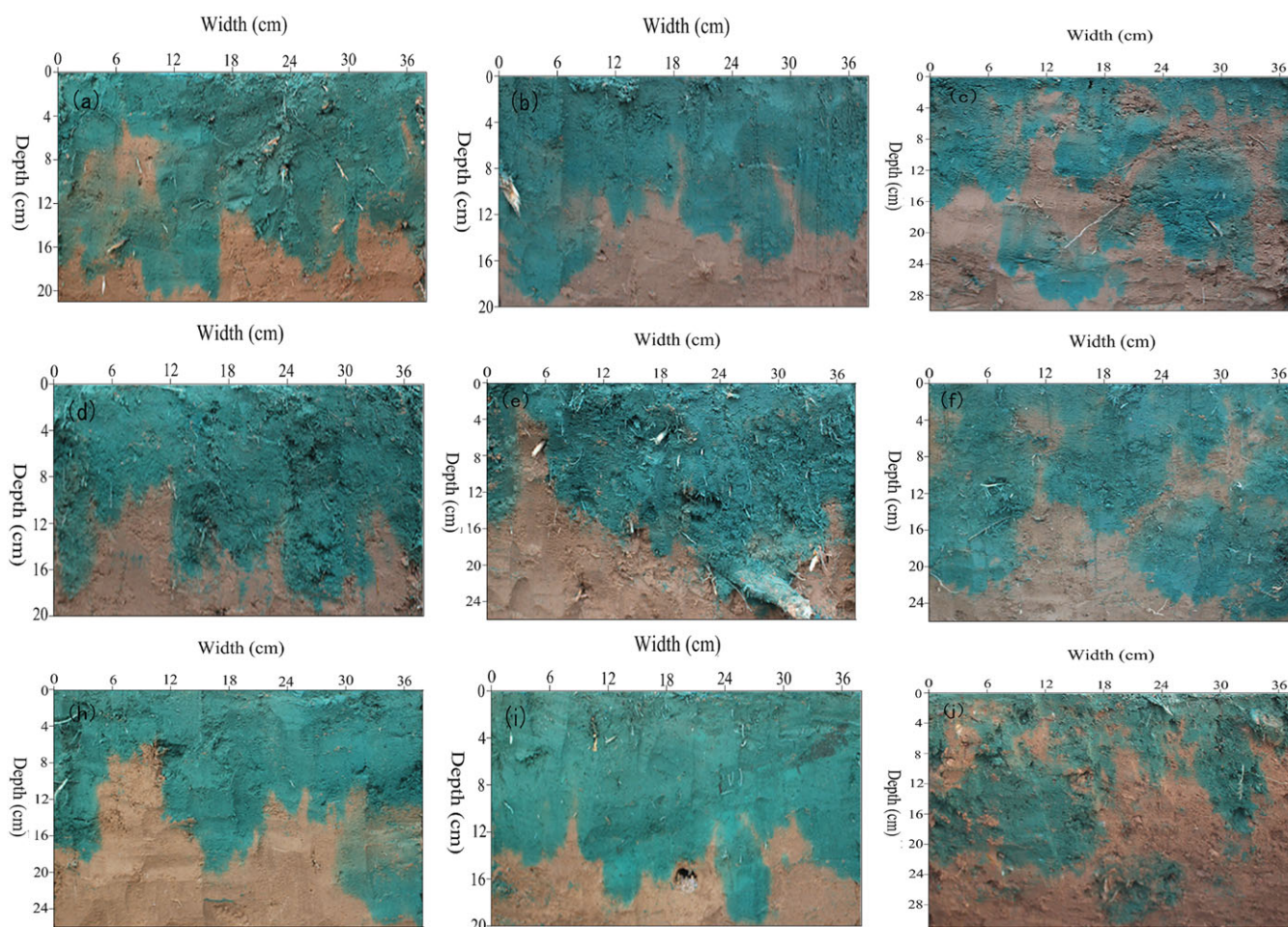


FIGURE 3 Example dye images for vertical soil sections in the three hillslopes. (a), (b), and (c) represent typical profile sections at the upslope, midslope, and downslope, respectively, in plantation slope; (d), (e), and (f) represent typical profile sections at upslope, midslope, and downslope, respectively, in natural forestland slope; and (h), (i), and (j) represent typical profile sections at the upslope, midslope, and downslope, respectively, in natural grassland slope [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 3 Preferential flow variables at the sample sites

Treatment		Infiltration time (min)	ID (cm)	UniFr (cm)	DC (%)	PF-fr (%)	LI	PIV (mm)	Contribution (%)
Transect 1									
PL	Upslope	16.62	22	16.9	47.98	11.95	128.16	5.23	19.23
	Midslope	24.92	20	11.5	34.23	16.01	179.47	7.79	33.90
	Downslope	45.57	38	14.4	55.95	35.65	381.84	21.73	56.64
NF	Upslope	20.60	26	13	45.86	29.11	197.89	12.41	39.21
	Midslope	24.35	26	15.3	43.74	12.56	208.68	7.10	31.97
	Downslope	24.90	26	10.1	50.73	50.23	275	12.08	51.62
NG	Upslope	20.47	20	13	37.63	13.62	141.58	5.21	23.46
	Midslope	50.77	24	13.3	44.22	24.81	213.42	8.59	37.53
	Downslope	120.01	30	11.2	45.32	51.85	271.58	18.79	52.26
Transect 2									0.00
PL	Upslope	20.20	21	14.8	42.15	12.22	174.47	6.47	24.04
	Midslope	28.01	24	9.9	39.49	37.32	260.26	14.24	49.35
	Downslope	40.47	30	5.0	46.82	26.70	355.26	27.80	74.32
NF	Upslope	21.48	20	12.2	38.90	21.59	155.53	7.66	31.36
	Midslope	23.60	24	7.5	29.44	36.31	258.68	13.85	45.16
	Downslope	24.75	24	16.1	44.66	9.87	189.94	5.06	31.99
NG	Upslope	30.25	20	7.3	37.90	38.21	164.74	7.80	43.11
	Midslope	55.53	26	11.4	40.26	29.22	187.37	17.14	50.31
	Downslope	88.53	31	6.2	43.83	64.64	370	30.07	75.21

Note. Contribution = the contribution of preferential infiltration to total infiltration; DC = dye coverage; ID = infiltration depth; LI = Length index; NF = natural forestland; NG = natural grassland; PF-fr = preferential flow fraction; PIV = preferential infiltration volume; PL = plantation; UniFr = uniform infiltration depth.

respectively) was larger than that in NF (51.62%, 31.99% in Transect 1, Transect 2, respectively). Similarly, the results from ID, DC, LI, and PIV showed that in the slope position, degree of preferential flow was larger in PL and NG than that in NF. However, the degree of preferential flow at the upper slope in NF was larger than that in PL and NG.

3.2 | Spatial variation of soil water infiltration

Infiltration isograms indicated the spatial distribution of water infiltration in the three vegetation types (Figure 4). These isograms accurately reflect the movement process of infiltrating water in the vertical profiles. In PL and NG, ID was greatest (greater than 30 cm) at the slope foot, and less than 26 cm on the upper slope. Different extents of preferential flow can cause diverse spatial distributions of soil water content (Jiang et al., 2015). A belt of high water infiltration generally appeared in upper slope. The belt was observed in the upper slope of PL at a depth of approximately 0–15 cm, which might represent the preferential flow pathway in the vertical direction. However, the belt of high water infiltration value primarily appeared in the shallow layer (soil depth of less than 10 cm) in the upper slope of NF and a shallower layer (soil depth of less than 8 cm) in the upper slope of NG. Additionally, scatters in the high water infiltration content appeared more frequently at the slope foot, which might represent lateral flow pathway.

Figure 5 presents the spatial distribution of infiltration volume with soil depth. The amount of water infiltration in the vertical profile directly reflects soil infiltrability in each layer and provides evidence for preferential flow. Generally, the infiltration volume was largest in the surface layer, where matrix flow was predominant. The infiltration volume decreased with increasing soil depth in the shallow layer, whereas in deeper soil layers (below UniFr), infiltration tended to increase with increasing soil depth, indicating preferential flow in these layers. The phenomenon frequently occurred at the downslope in PL, NF, and NG.

The spatial variation in change in soil water content in vertical was analyzed using semivariogram, and the results are shown in Table 4. The mean spatial autocorrelation range (A) values at the downslope in PL and NG were higher than that in NF, whereas the A values at the upper slope in PL and NG were lower than that in NF. The spatial class at each site was strong.

3.3 | Impact factors of preferential flow

The contributions of environmental variables to preferential flow can be evaluated by examining the correlation coefficients between preferential flow and environmental variables. The values of the environmental variables, that is, BD, Ks, ISWC, RFC, slope position, clay, silt, and sand, at each site are listed in Table 1. Table 5 presents the correlation coefficients of preferential flow and the environmental variables. Generally, Ks, ISWC, RFC, and sand had negative correlation with UniFr and positive correlation with other preferential indexes. BD, slope position, clay, and sand exhibited positive correlation with UniFr and negative correlation with other preferential indexes. In PL, the relationship between BD, Ks, RFC, slope position, clay, silt, sand and LI, PIV, contribution was significantly correlated; in NF, Ks was significantly correlated with LI, PIV, and contribution; and in NG, Ks, ISWC, RFC, slope position was significant positively correlated with ID, LI, PIV, and contribution.

4 | DISCUSSION

4.1 | The contribution of preferential flow and matrix flow

Dye tracers offer visual information on water infiltration along preferential flow pathways that bypass the soil matrix (Flury & Wai, 2003).

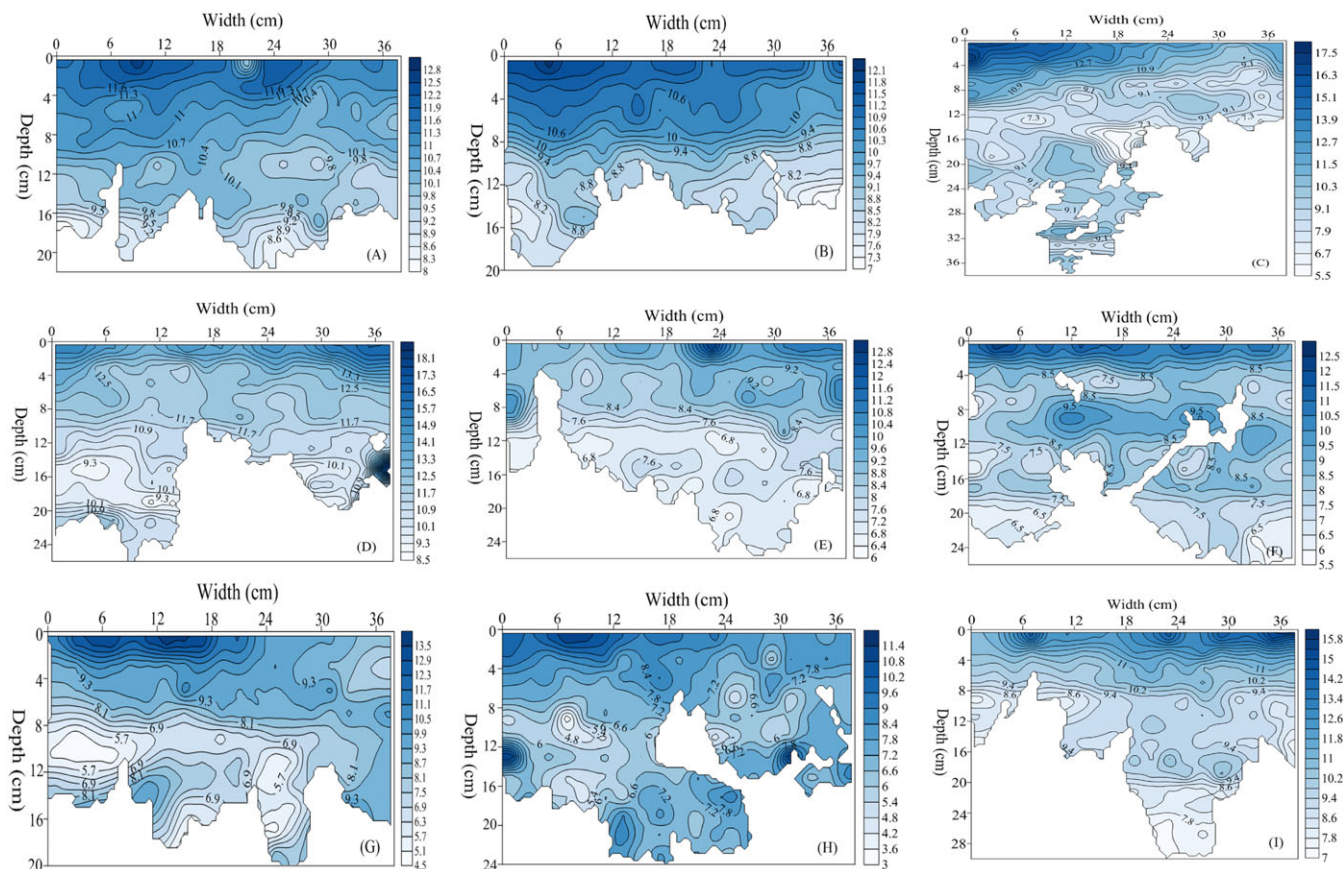


FIGURE 4 Contour maps of water infiltration in vertical profiles in plantation (PL), natural forestland (NF), and natural grassland (NG); (A)–(I) represent Transect 1; (a)–(i) represent Transect 2; (A)(a), (B)(b), (C)(c) represent at the upslope, midslope, downslope, respectively, in PL; (D)(d), (E)(e), (F)(f) represent at the upslope, midslope, downslope, respectively, in NF; (G)(g), (H)(h), (I)(i) represent at the upslope, midslope, downslope, respectively, in NG [Colour figure can be viewed at wileyonlinelibrary.com]

Commonly used preferential flow indexes (e.g., ID, UniFr, DC, PF-fr, and LI) based on dye images can only qualitatively evaluate the extent of preferential flow. In our study, dye tracer experiments with water movement to measure soil pore morphology and water infiltration, and new indexes (e.g., PIV and its contribution to total infiltration) proposed were effective for distinguishing the contribution of preferential flow from matrix flow, which can better provide information on soil infiltration.

In the Loess Plateau, low rainfall and high evaporation lead to low soil moisture levels that barely meet the needs for vegetation growth (Yang, Wei, Chen, & Mo, 2012). In addition, little groundwater can be used by plants. Thus, improper land-use types could lead to soil desiccation and land degradation. In fact, stand origin types had a profound effect on the occurrence of preferential flow, which is responsible for the depth and heterogeneity of water infiltration by changing soil properties (e.g., soil water content and soil structure) and forming root channels. Among the three vegetation types, PL and NG at the downslope exhibited higher contribution of preferential flow to total infiltration than NF (Table 3) due to the presence of RFC in PL and NG, whereas PL and NG at the upper slope had lower contribution of preferential flow than NF, indicating NF had a higher degree of preferential flow than PL and NG without effect of RFC. These results revealed that at the upper slope, the role of root systems in the formation of preferential flow pathway is emphasized. Water infiltrates into soil

via networks of large pore and voids that are mainly formed by root systems. The root systems of *R. pseudoacacia* were deeply spread, whereas in NF, many shrubs and woods make abundant roots distribute in the upper soil layers. The plant roots can change soil properties by reducing soil BD and increasing total soil porosity. Improved soil properties can enhance soil development, which is vitally critical in the soil hydrological process (e.g., soil water flow) and soil water infiltration rates (Wang, Fan, Cao, & Liang, 2016). These results that NF has larger degree of preferential flow than NG agreed with those obtained by Alaoui et al. (2011), who indicated that macropores were more efficient at transporting water vertically downward in forest than those in grassland soil. However, considering all the preferential flow indexes, PL did not exhibit greatly larger extent of preferential flow than NG, conflict with Alaoui et al. (2011). A potential explanation may be due to the location of the previous study in Innerrüthen where the mean annual temperature is 5.9 °C, and the total annual precipitation is 1,274 mm. Thus, the sufficient precipitation and the low amount of evapotranspiration led soil water content similar in forestland and grassland. However, in the Loess Plateau, where the annual average precipitation is only 579.5 mm, artificial forest systems have experienced water deficits, with significantly lower soil water content than grassland (Table 1). Higher ISWC significantly related to increased preferential flow in NG (Table 5) evidence for the results. Regarding slope position, downslope in PL (56.64%,

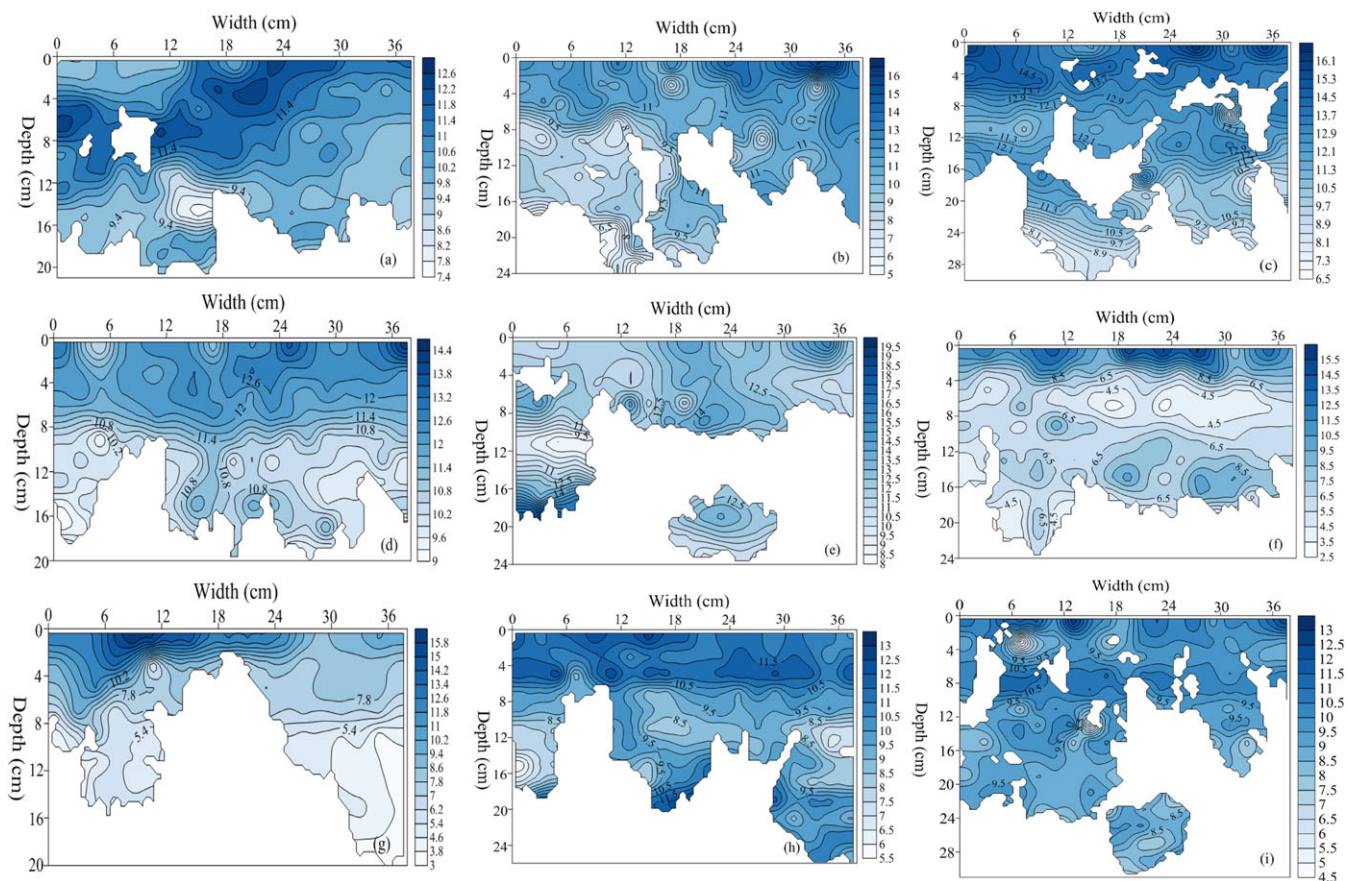


FIGURE 4 Continued.

74.32%) and NG (52.26%, 75.21%) made greater contributions to total infiltration than upslope or midslope did, but not in NF, indicating that RFC plays a bigger role in the formation of preferential flow.

Dye transport is slower than water infiltration (Germán-Heins & Flury, 2000); therefore, the water infiltration distribution reveals preferential flow pathways and infiltration pattern more accurately than the distribution of dye (Jiang et al., 2015). Thus, preferential flow pathways can be more clearly observed, as illustrated in Figure 4. The belt of high water infiltration might result from the largely uniform soil structure. Several scattered areas of high water infiltration at the downslope might be caused by the nonuniform distributions of rock fragments (Table 1), which created potential pathways for the macropore-matrix exchange of water in the soil. Additionally, the spatial variation of water infiltration content, as determined by using geostatistical methods, was related to the A value. Infiltration patterns with high degrees of spatial variation exhibit high A values (Jiang et al., 2015; Zuo et al., 2008), indicating that the spatial variability at the downslope in PL and NG was higher than that in NF, adversely, the variability at the upper slope in PL and NG was lower than that in NF. Spatial class indicates strong spatial dependence at nugget percentages less than 25% (Iqbal, Thomasson, Jenkins, Owens, & Whisler, 2005), which demonstrates that infiltration patterns have strong spatial dependence within each stand origin type and slope position. Spatial variation of soil water content can be caused by water transfer between preferential flow paths and the surrounding soil matrix different from depending on the soil texture (Beven & Germann, 1982). Water transfer was higher between preferential flow paths and the

surrounding soil matrix. However, soil water in the surrounding soil matrix transferred to the surrounding environment is difficult, because the soil matrix stored more capillary water was compacted (Zhang et al., 2017). The root systems or rock fragments would enhance preferential flow in the soil and increase spatial variation in water infiltration content. Thus, increasing degree of preferential flow might increase the spatial heterogeneity of water infiltration in vertical. The results were consistent with those of Shinohara and Otsuki (2015), who found that the spatial variation of SWC was higher in a bamboo stand than in a broadleaved stand, and well-developed preferential flow paths in the soil profiles of the bamboo stand might predominate.

4.2 | The main factors impacting preferential flow

Preferential flow is strongly affected by factors such as vegetation type, soil structure, ISWC, and RFC. Zhang et al. (2016) noted that cracks formed by rock fragments could provide preferential pathways. In addition, Dadkhah and Gifford (1980) indicated that RFC had a positive effect on water infiltration in noncompacted soils. In our study, RFC can promote lateral flow and increase the amount of water infiltration, as evidenced by lateral flow and variable infiltration patterns dominant at the downslope in PL and NG. An interphase area between rock fragments and the surrounding fine earth exists of high macroporosity, which provides pathway for lateral flow and affects the flow process, because the space among rock fragments is hardly completely filled in by the fine phase, or larger rock fragments prevent the surrounding soil matrix from packing (Zhang et al., 2016). The

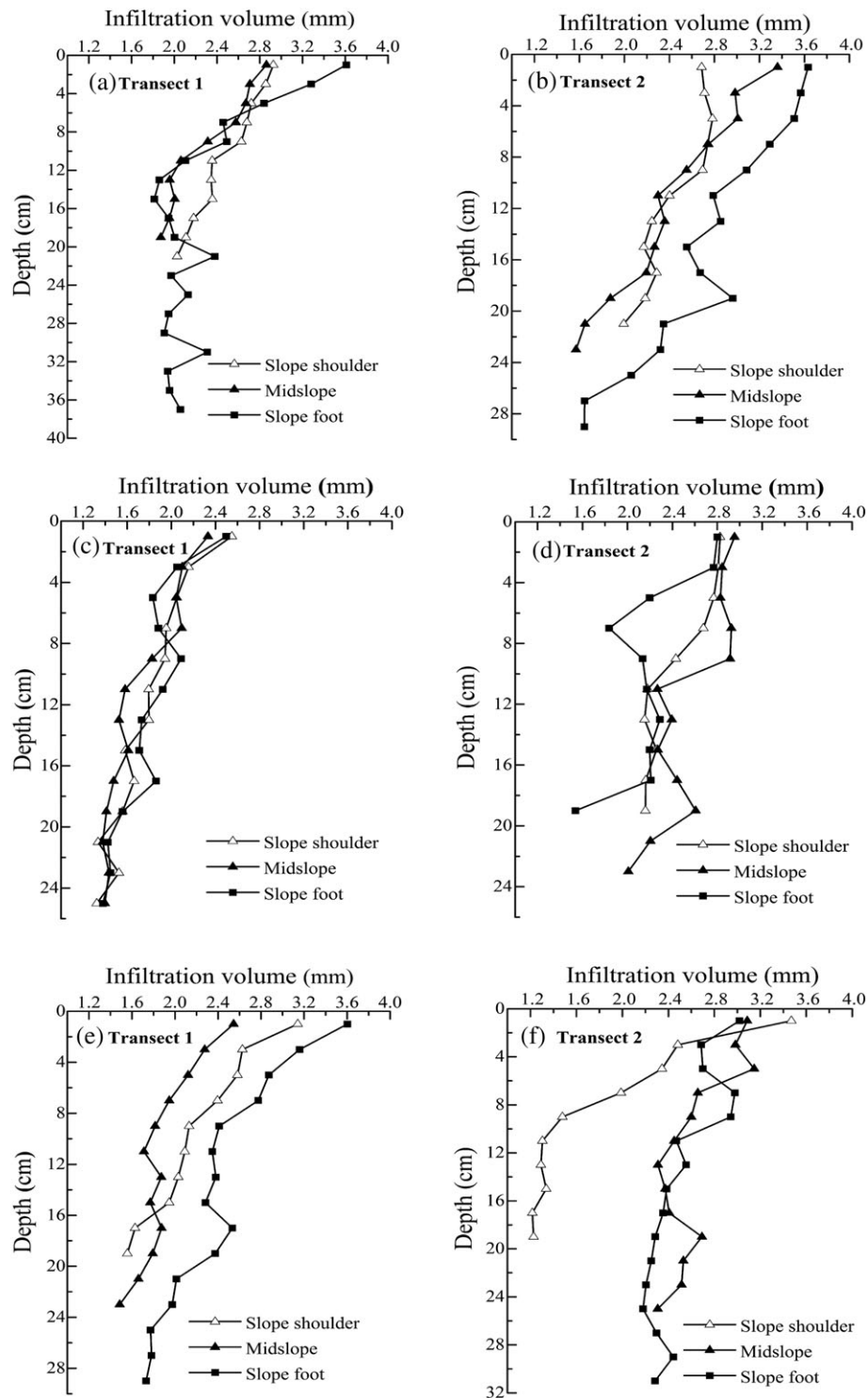


FIGURE 5 Variation in water infiltration volume with soil depth in the vertical profiles in three vegetation types. (a), (b): plantation; (c), (d): natural forestland; (e), (f): natural grassland

effects of volumetric content of rock fragments on water movement are exerted by affecting the radii, density and area of macropores, and connectivity among the macropores to some extent (Shi et al., 2012). Rock fragments, which were randomly distributed in the present study, have distinct effects on soil hydrological processes; thus, RFC affected the spatial variation of infiltration patterns. Lateral flow plays an important role in hydrological processes, particularly water movement along the slope toward streams and catchment outlets (Laine-Kaulio et al., 2015). This observation may be related to

assumptions that downslope position on the Loess Plateau has effect on preferential flow and base flow in the gully originated from lateral flow at the downslope in PL and NG due to presence of RFC.

ISWC had a significantly positive relationship with degree of preferential flow (ID, DC, PF-fr, and PIV) in NG (Table 5), indicating higher ISWC can motivate preferential flow pathway. Our results conflict with those of Wu et al. (2015) and Hardie et al. (2011), who found that a high ISWC could restrict preferential flow in the field. However, consistent with our results, Kung et al. (2000) found that increased

TABLE 4 Parameters for the theoretical semivariograms of change in soil water content in the three vegetation types

Treatment		Model	A (cm)	C ₀	(C ₀ + C)	C ₀ /(C ₀ + C) (%)	Spatial class	R ²	RSS
Transect1									
PL	Upslope	Ga	15.18	0.001	0.071	2.12	S	.98	1.49E-04
	Midslope	Ga	24.16	0.003	0.114	2.82	S	.99	2.39E-05
	Downslope	Ga	32.16	0.018	0.296	6.19	S	.98	1.65E-03
NF	Upslope	Sp	26.70	0.001	0.123	1.06	S	.97	3.78E-04
	Midslope	Ex	28.01	0.0008	0.142	0.56	S	.98	1.62E-04
	Downslope	Ga	28.67	0.009	0.343	2.62	S	.99	9.74E-04
NG	Upslope	Ex	15.18	0.0001	0.051	0.19	S	.96	1.30E-04
	Midslope	Ex	26.71	0.0001	0.058	0.17	S	.93	2.62E-04
	Downslope	Li	42.87	0.001	0.936	0.11	S	.97	9.88E-03
Transect 2									
PL	Upslope	Sp	22.63	0.0003	0.132	0.23	S	.99	2.78E-04
	Midslope	Ga	26.93	0.020	0.181	10.82	S	.99	9.09E-05
	Downslope	Ga	42.11	0.028	0.308	9.25	S	.96	1.71E-03
NF	Upslope	Sp	28.56	0.0001	0.040	0.25	S	.97	5.41E-05
	Midslope	Ex	29.64	0.0001	0.181	0.06	S	.98	7.53E-04
	Downslope	Ex	21.09	0.001	0.332	0.30	S	.82	0.0254
NG	Upslope	Ex	12.80	0.0001	0.039	0.25	S	.92	1.37E-04
	Midslope	Ex	24.18	0.001	0.291	0.34	S	.97	1.19E-03
	Downslope	Li	48.45	0.0001	0.289	0.03	S	.98	6.27E-04

Note. A = spatial autocorrelation range; C₀ = nugget variance; (C₀ + C) = sill; Ex = exponential model; Ga = Gaussian model; Li = Line model; NF = natural forestland; NG = natural grassland; PL = plantation; RSS = residual sum of squares; S = strong spatial dependence; Sp = spherical model.

TABLE 5 Correlations coefficients of preferential flow and environmental variables

Preferential flow variable		BD (g·cm ⁻³)	K _s	ISWC (g·g ⁻¹)	RFC (cm ³ ·cm ⁻³)	Slope position	Clay (%)	Silt (%)	Sand (%)
PL	ID	-0.872*	0.877*	0.001	.928**	-0.71	-0.639	-0.672	0.674
	UniFr	0.585	-0.577	0.136	-0.398	0.68	0.820*	0.752	-0.758
	DC	-0.515	0.529	0.294	0.731	-0.221	-0.189	-0.352	0.346
	PF-fr	-0.684	0.677	0.095	0.54	-0.769	-0.743	-0.464	0.479
	LI	-0.955**	0.956**	-0.028	0.912*	-0.880*	-0.860*	-0.826*	0.831*
	PIV	-0.949**	0.950**	-0.011	0.906*	-0.858*	-0.912*	-0.928**	0.932**
	Contribution	-0.924**	0.921**	-0.078	0.817*	-0.906*	-.965**	-0.915*	0.921**
NF	ID	-0.442	0.421	0.224	-	-0.417	-0.022	-0.25	0.225
	UniFr	0.056	-0.017	0.447	-	-0.024	0.583	0.525	-0.537
	DC	0.017	0.587	0.398	-	-0.206	0.817*	0.517	-0.558
	LI	-0.064	0.811*	-0.254	-	-0.115	-0.158	-0.291	0.278
	PF-fr	-0.651	0.617	0.166	-	-0.611	-0.491	-0.714	0.694
	PIV	0.075	0.843*	-0.562	-	0.149	-0.416	-0.385	0.393
	Contribution	-0.301	0.839*	-0.059	-	-0.341	-0.289	-0.49	0.471
NG	ID	0.027	0.918**	0.985**	0.867*	-0.966**	0.041	-0.754	0.734
	UniFr	-0.509	-0.402	-0.205	-0.427	0.118	0.028	0.451	-0.442
	DC	-0.164	0.764	0.892*	0.704	-.918**	0.449	-0.352	0.325
	LI	0.417	0.876*	0.763	0.872*	-0.682	0.04	-0.749	0.729
	PF-fr	0.225	0.908*	0.903*	0.873*	-0.833*	0.256	-0.614	0.589
	PIV	0.144	0.847*	0.863*	0.806	-0.818*	-0.024	-0.773	0.756
	Contribution	0.137	0.778	0.771	0.738	-0.748	0.085	-0.695	0.675

Note. BD = bulk density; DC = dye coverage; ID = infiltration depth; ISWC = initial soil water content; K_s = soil saturated hydraulic conductivity; LI = Length index; NF = natural forestland; NG = natural grassland; PF-fr = preferential flow fraction; PIV = preferential infiltration volume; PL = plantation; RFC = rock fragments coverage; UniFr = uniform infiltration depth.

*Represents significant correlation ($p < .05$).

**Represents significant correlation ($p < .01$).

preferential flow was related to increased initial soil moisture. Sanders, Abou Najm, Mohtar, Kladienko, and Schulze (2012) also indicated that drier ISWC conditions can result in a higher sorptive capacity of the soil matrix. A potential explanation for our results is that preferential flow occurred in stable macropore or biopore (rather than shrinkage cracks) pathways in clay loam or sandy clay loam soils, where a higher ISWC can increase infiltration depth of macropore flow as well as total infiltration volume as it reduce the lateral flow from macropores into the soil matrix (Hardie et al., 2011). Hence, in the Loess Plateau, the

extent of preferential flow observed in NG was related to the slope's higher ISWC.

K_s also exhibited a strong positive correlation with preferential flow. This finding is consistent with that of Laine-Kaulio et al. (2015), who demonstrated that DC showed a second order polynomial increase in relation to K_s. Additionally, strong correlation was found between BD and preferential flow in PL, presumably in relation to soil porosity. The relationship between particle size and preferential flow revealed that in clay loam or sandy clay loam soil, higher clay and silt

content tended to promote matrix flow and that higher sand content was prone to preferential flow. The results are in agreement with those of Zhang et al. (2015), who found that sandy soil can contribute to macropore density, and those of Alaoui et al. (2011), who found that the higher clay content in the topsoil layers of the hillslopes tended to generate runoff. It is possible that clay and silt promoted matrix flow and consequently delayed water routing into macropores by sorptivity, causing low preferential flow.

5 | CONCLUSIONS

The preferential flow and infiltration patterns in PL, NF, and NG slopes were accurately represented by dye tracer experiment with soil water movement. PIV and contribution of preferential infiltration can effectively evaluate preferential flow. In addition, among the two stand origins, at the upslope and midslope, NF has higher degree of preferential flow that role of macropore flow was dominant, contribution of preferential flow, and spatial variability of water infiltration than PL, because the presence of abundant root systems in upper soil of NF plays a critical role in preferential flow by root channels and changing soil physical properties. At the downslope, lateral flow was dominant in downslope in PL and NG, due to the presence of RFC. RFC also made contribution of preferential flow and spatial variability of water infiltration at the downslope in PL higher than that in NF. And soil properties greatly affected degree of preferential flow; in PL, BD, Ks, clay, silt, and sand significantly correlated with preferential flow; in NF, Ks, and clay significantly influenced the degree; and in NG, Ks, and ISWC significantly related to the degree. Thus, NF has larger degree of preferential flow in hydrological process in upper soil that can effectively increase preferential infiltration in upper soil layer, replenish deep soil water, and reduce runoff and land degradation. The presence of RFC made lateral flow dominant at the downslope on the Loess hillslope, which was a potentially important source of base flow in gullies. The important preferential flow functions should be considered when studying land degradation, as well as overland flow and surface soil erosion.

ACKNOWLEDGMENTS

This research was financially supported by The National Key Research and Development Program of China (2016YFC0501704), and the National Natural Science Foundation of China (51779004 and 41530858). The authors thank Du Shengli and Tian Zheng for helping with the field work, and American Journal Experts for English language editing.

ORCID

Xuemei Mei  <http://orcid.org/0000-0003-1380-0149>

REFERENCES

- Alaoui, A., Caduff, U., Gerke, H., Gerkeb, H., & Weingartner, R. (2011). A preferential flow effects on infiltration and runoff in grassland and forest soils. *Vadose Zone Journal*, 10, 367–377. <https://doi.org/10.2136/vzj2010.0076>
- Allaire, S. E., Bochove, E. V., Denault, J. T., Dadfar, H., Thériault, G., Charles, A., & Jong, R. D. (2011). Preferential pathways of phosphorus movement from agricultural land to water bodies in the Canadian Great Lakes basin: A predictive tool. *Canadian Journal of Soil Science*, 91, 361–374. <https://doi.org/10.4141/CJSS09121>
- Allaire, S. E., Roulier, S., & Cessna, A. J. (2009). Quantifying preferential flow in soils: A review of different techniques. *Journal of Hydrology*, 378, 179–204. <https://doi.org/10.1016/j.jhydrol.2009.08.013>
- Bachmair, S., Weiler, M., & Nutzmann, G. (2009). Controls of land use and soil structure on water movement: Lessons for pollutant transfer through the unsaturated zone. *Journal of Hydrology*, 369, 241–252. <https://doi.org/10.1016/j.jhydrol.2009.02.031>
- Bargués Tobella, A., Reese, H., Almaw, A., Bayala, J., Malmer, A., Laudon, H., & Ilstedt, U. (2014). The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. *Water Resources Research*, 50, 3342–3354. <https://doi.org/10.1002/2013WR015197>
- Beven, K., & Germann, P. (1982). Macropores and water flow in soils. *Water Resources Research*, 18, 1311–1325. <https://doi.org/10.1029/WR018i005p01311>
- Bogner, C., Wolf, B., Schlather, M., & Huwe, B. (2008). Analysing flow patterns from dye tracer experiments in a forest soil using extreme value statistics. *European Journal of Soil Science*, 59, 103–113. <https://doi.org/10.1111/j.1365-2389.2007.00974.x>
- Bouma, J., & De Laat, P. J. M. (1981). Estimation of the moisture supply capacity of some swelling clay soils in the Netherlands. *Journal of Hydrology*, 49, 247–259. [https://doi.org/10.1016/0022-1694\(81\)90216-X](https://doi.org/10.1016/0022-1694(81)90216-X)
- Cey E. E. 2007. Macropore flow and transport dynamics in partially saturated low permeability soils. PhD thesis, University of Waterloo, Waterloo, Ontario, Canada.
- Chen, L. D., Huang, Z. L., Gong, J., Fu, B. J., & Huang, Y. L. (2007). The effect of land cover/vegetation on soil water dynamic in the hilly area of the Loess Plateau, China. *Catena*, 70, 200–208. <https://doi.org/10.1016/j.catena.2006.08.007>
- Dadkhah, M., & Gifford, G. F. (1980). Influence of vegetation, rock cover and trampling on infiltration rates and sediment production. *Journal of the American Water Resources Association*, 16, 979–986. <https://doi.org/10.1111/j.1752-1688.1980.tb02537.x>
- Flury, M., & Flühler, H. (1994). Brilliant blue FCF as a dye tracer for solute transport studies—a toxicological overview. *Journal of Environmental Quality*, 23, 1108–1112. <https://doi.org/10.2134/jeq1994.00472425002300050037x>
- Flury, M., & Wai, N. N. (2003). Dyes as tracers for vadose zone hydrology. *Reviews of Geophysics*, 41, 1002. <https://doi.org/10.1029/2001RG000109>
- Gao, X., Zhao, X., Wu, P., Brocca, L., & Zhang, B. (2016). Effects of large gullies on catchment-scale soil moisture spatial behaviors: A case study on the Loess Plateau of China. *Geoderma*, 261, 1–10. <https://doi.org/10.1016/j.geoderma.2015.07.001>
- Gerke, H. H. (2006). Preferential flow description for structured soils. *Journal of Plant Nutrition and Soil Science*, 169, 382–400. <https://doi.org/10.1002/jpln.200521955>
- Germán-Heins, J., & Flury, M. (2000). Sorption of Brilliant Blue FCF in soils as affected by pH and ionic strength. *Geoderma*, 97, 87–101. [https://doi.org/10.1016/S0016-7061\(00\)00027-6](https://doi.org/10.1016/S0016-7061(00)00027-6)
- Gjettermann, B., Nielsen, K. L., Petersen, C. T., Jensen, H. E., & Hansen, S. (1997). Preferential flow in sandy loam soils as affected by irrigation intensity. *Soil Technology*, 11, 139–152. [https://doi.org/10.1016/S0933-3630\(97\)00001-9](https://doi.org/10.1016/S0933-3630(97)00001-9)
- Hagedorn, F., & Bundt, M. (2002). The age of preferential flow paths. *Geoderma*, 108, 119–132. [https://doi.org/10.1016/S0016-7061\(02\)00129-5](https://doi.org/10.1016/S0016-7061(02)00129-5)
- Hardie, M. A., Cotching, W. E., Doyle, R. B., Holz, G., Lisson, S., & Mattern, K. (2011). Effect of antecedent soil moisture on preferential flow in a texture-contrast soil. *Journal of Hydrology*, 398, 191–201. <https://doi.org/10.1016/j.jhydrol.2010.12.008>

- Holden, J., & Burt, T. P. (2003). Runoff production in blanket peat covered catchments. *Water Resources Research*, 39. <https://doi.org/10.1029/2002WR001956>
- Hu, W., Shao, M. G., Han, F. P., & Reichardt, K. (2011). Spatio-temporal variability behavior of land surface soil water content in shrub- and grass-land. *Geoderma*, 162, 260–272. <https://doi.org/10.1016/j.geoderma.2011.02.008>
- Iqbal, J., Thomasson, J. A., Jenkins, J. N., Owens, P. R., & Whisler, F. D. (2005). Spatial variability analysis of soil physical properties of alluvial soils. *Soil Science Society of America Journal*, 69, 1338–1350. <https://doi.org/10.2136/sssaj2004.0154>
- Jiang, W. Y., Cheng, Y. F., Yang, X. X., & Yang, S. L. (2013). Chinese Loess Plateau vegetation since the Last Glacial Maximum and its implications for vegetation restoration. *Journal of Applied Ecology*, 50, 440–448. <https://doi.org/10.1111/1365-2664.12052>
- Jiang, X. J., Liu, S., & Zhang, H. (2017). Effects of different management practices on vertical soil water flow patterns in the Loess Plateau. *Soil and Tillage Research*, 166, 33–42. <https://doi.org/10.1016/j.still.2016.10.001>
- Jiang, X. J., Liu, X. E., Wang, E. H., Li, X. G., Sun, R., & Shi, W. J. (2015). Effects of tillage pan on soil water distribution in alfalfa-corn crop rotation systems using a dye tracer and geostatistical methods. *Soil & Tillage Research*, 150, 68–77. <https://doi.org/10.1016/j.still.2015.01.009>
- Klute, A., & Dirksen, C. (1986). Hydraulic conductivity of saturated soils. In A. Klute (Ed.), *Methods of Soil Analysis* (pp. 694–700). Madison, Wisconsin, USA: ASA & SSSA.
- Kung, K. J. S., Steenhuis, T. S., Klavivko, E. J., Gish, T. J., Bubenzer, G., & Helling, C. S. (2000). Impact of preferential flow on the transport of adsorbing and non-adsorbing tracers. *Soil Science Society of America Journal*, 64, 1290–1296. <https://doi.org/10.2136/sssaj%202000.6441290x>
- Laine-Kaulio, H., Backnas, S., Koivusalo, H., & Lauren, A. (2015). Dye tracer visualization of flow patterns and pathways in glacial sandy till at a boreal forest hillslope. *Geoderma*, 259, 23–34. <https://doi.org/10.1016/j.geoderma.2015.05.004>
- Ritsema, C. J., & Dekker, L. W. (1996). Influence of sampling strategy on detecting preferential flow paths in water-repellent sand. *Journal of Hydrology*, 177, 33–45. [https://doi.org/10.1016/0022-1694%20\(95\)02795-5](https://doi.org/10.1016/0022-1694%20(95)02795-5)
- Sanders, E. C., Abou Najm, M. R., Mohtar, R. H., Klavivko, E., & Schulze, D. (2012). Field method for separating the contribution of surface-connected preferential flow pathways from flow through the soil matrix. *Water Resources Research*, 48. <https://doi.org/10.1029/2011WR011103>
- Shi, Z. J., Xu, L. H., Wang, Y. H., Yang, X. H., Jia, Z. Q., Guo, H., ... Yu, P. T. (2012). Effect of rock fragments on macropores and water effluent in a forest soil in the stony mountains of the Loess Plateau, China. *African Journal of Biotechnology*, 11, 9350–9393. <https://doi.org/10.5897/AJB12.145>
- Shinohara, Y., & Otsuki, K. (2015). Comparisons of soil-water content between a Moso bamboo (*Phyllostachys pubescens*) forest and an evergreen broadleaved forest in western Japan. *Plant Species Biology*, 30, 96–103. <https://doi.org/10.1111/1442-1984.12076>
- van Schaik, N. L. M. B. (2009). Spatial variability of infiltration patterns related to site characteristics in a semi-arid watershed. *Catena*, 78, 36–47. <https://doi.org/10.1016/j.catena.2009.02.017>
- Wang, L., Wang, Q. J., Wei, S. P., Shao, M. A., & Li, Y. (2008). Soil desiccation for Loess soils on natural and regrown areas. *Forest Ecology Management*, 255, 2467–2477. <https://doi.org/10.1016/j.foreco.2008.01.006>
- Wang, X. P., Li, X. R., Xiao, H. L., Berndtsson, R., & Pan, Y. X. (2007). Effects of surface characteristics on infiltration patterns in an arid shrub desert. *Hydrological Processes*, 21, 72–79. <https://doi.org/10.1002/hyp.6185>
- Wang, Y., Fan, J., Cao, L., & Liang, Y. (2016). Infiltration and runoff generation under various cropping patterns in the red soil region of China. *Land Degradation & Development*, 27, 83–91. <https://doi.org/10.1002/ldr.2460>
- Weiler, M., & Flühler, H. (2004). Inferring flow types from dye patterns in macroporous soils. *Geoderma*, 120, 137–153. <https://doi.org/10.1016/j.geoderma.2003.08.014>
- Wu, Q. H., Liu, C. L., Lin, W. J., Zhang, M., Wang, G. L., & Zhang, F. W. (2015). Quantifying the preferential flow by dye tracer in the North China Plain. *Journal of Earth Science*, 26, 435–444. <https://doi.org/10.1007/s12583-014-0489-4>
- Yang, L., Wei, W., Chen, L. D., & Mo, B. R. (2012). Response of deep soil moisture to land use and afforestation in the semi-arid Loess Plateau, China. *Journal of Hydrology*, 475, 111–122. <https://doi.org/10.1016/j.jhydrol.2012.09.041>
- Zhang, Y. H., Niu, J. Z., Yu, X. X., Zhu, W. L., & Du, X. Q. (2015). Effects of fine root length density and root biomass on soil preferential flow in forest ecosystems. *Forest Systems*, 24, 012. <https://doi.org/10.5424/fs/2015241-06048>
- Zhang, Y. H., Niu, J. Z., Zhang, M. X., & Zhu, W. L. (2017). Interaction between plant roots and soil water flow in response to preferential flow paths in Northern China. *Land Degradation & Development*, 28, 648–663. <https://doi.org/10.1002/ldr.2592>
- Zhang, Y. H., Zhang, M. X., Niu, J. Z., Li, H. L., Xiao, R., Zheng, H. J., & Bechd, J. (2016). Rock fragments and soil hydrological processes: Significance and progress. *Catena*, 147, 153–166. <https://doi.org/10.1016/j.catena.2016.07.012>
- Zuo, X. A., Zhao, H. L., Zhao, X. Y., Zhang, T. H., Guo, Y. R., Wang, S. K., & Drake, S. (2008). Spatial pattern and heterogeneity of soil properties in sand dunes under grazing and restoration in Horqin Sandy Land Northern China. *Soil & Tillage Research*, 99, 202–212. <https://doi.org/10.1016/j.still.2008.02.008>

How to cite this article: Mei X, Zhu Q, Ma L, Zhang D, Wang Y, Hao W. Effect of stand origin and slope position on infiltration pattern and preferential flow on a Loess hillslope. *Land Degrad Dev*. 2018;29:1353–1365. <https://doi.org/10.1002/ldr.2928>